

## Science

# General relativity: How Einstein's theory explains the universe, and more

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**A century ago, physicist Albert Einstein unveiled a theory that would change the world — general relativity.**

It would cement his place at the top of the pantheon of scientific minds, and see him transformed into a modern icon.

But what is general relativity, and why does it matter?

Ask someone on the street about Einstein and they'll likely say the word genius, maybe rattle off [E=mc<sup>2</sup>](#), and possibly talk about the nuclear bomb. But ask them to explain general relativity and things are likely to get a little quiet.

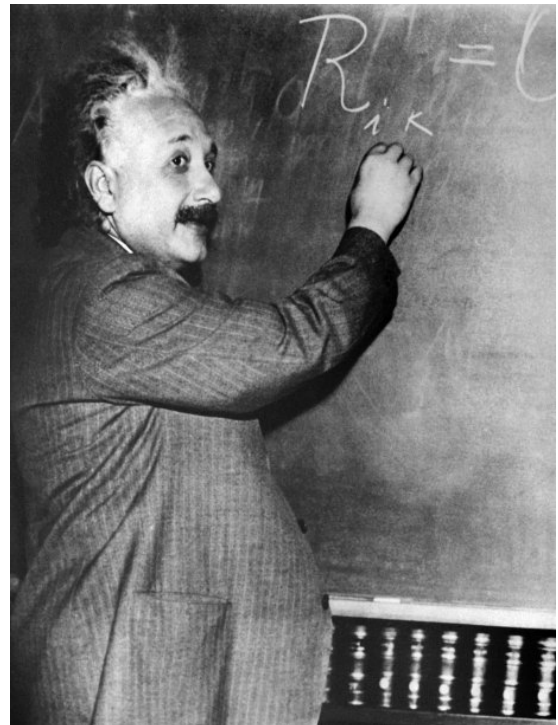
And for good reason. General relativity is full of mind-bending concepts like [warped spacetime and time dilation](#).

But it is really just Einstein's thorough explanation of what causes gravity, and how gravity affects matter, light and time.

Einstein had an issue with gravity. His thinking went like this. If you're in a stationary elevator and you drop a ball, the ball will fall to the floor of the lift at 9.8 metres per second per second — that's the rate of gravitational acceleration for everything that falls to Earth.

But exactly the same thing would happen if you were in a rocket that was accelerating upwards through space at 9.8 metres per second per second. Drop the ball and it will fall just like it would on Earth. That uniform acceleration has exactly the same effect as gravity.

And the opposite is true too. If your rocket was completely still in the middle of space, you and the ball would be weightless. And that would feel identical to the freefall you and the ball would go into on Earth if



[Albert Einstein's theory of relativity has been explaining how the universe works for the last 100 years](#) (AFP)

the elevator cables snapped.

The idea that there's no difference between the effect of gravity and the effect of uniform acceleration became known as the equivalence principle. And together with the idea of spacetime, it's the basis of Einstein's take on gravity — his theory of general relativity.

## Space and time

In November 1915, Einstein described gravity with the 10 heavy-handed field equations of general relativity, but what those equations basically say is — gravity is what you get whenever spacetime is warped, bent or stretched.

Spacetime isn't something we normally talk about, because we think of space (up-down, left-right, forwards-backwards) as separate from time. But they're not. Space and time are intimately connected: if you bend space, you affect time too. And bending/warping or stretching spacetime is the cause of all gravity in our universe.

The bending/warping is caused by matter, or anything with mass. In fact, every bit of matter (including you) is warping the spacetime around it, creating its own bit of gravity.

General relativity lets us calculate and predict exactly how anything with mass will warp spacetime, and how the gravity caused by that warped spacetime will affect not just the matter, but the space, time and light around it.

For tiny things like us, the warpage (yep that's a word) is miniscule, so Newton's 17th century law of gravity ( $F=Gm_1m_2/r^2$ ) is perfectly fine.

But massive things like stars warp and stretch space so much that Newton's law just doesn't work properly near stars – it only holds in "flat" space. At these massive scales, space, time and light are noticeably affected.

Even a middling-sized planet like Earth warps space enough to affect time.

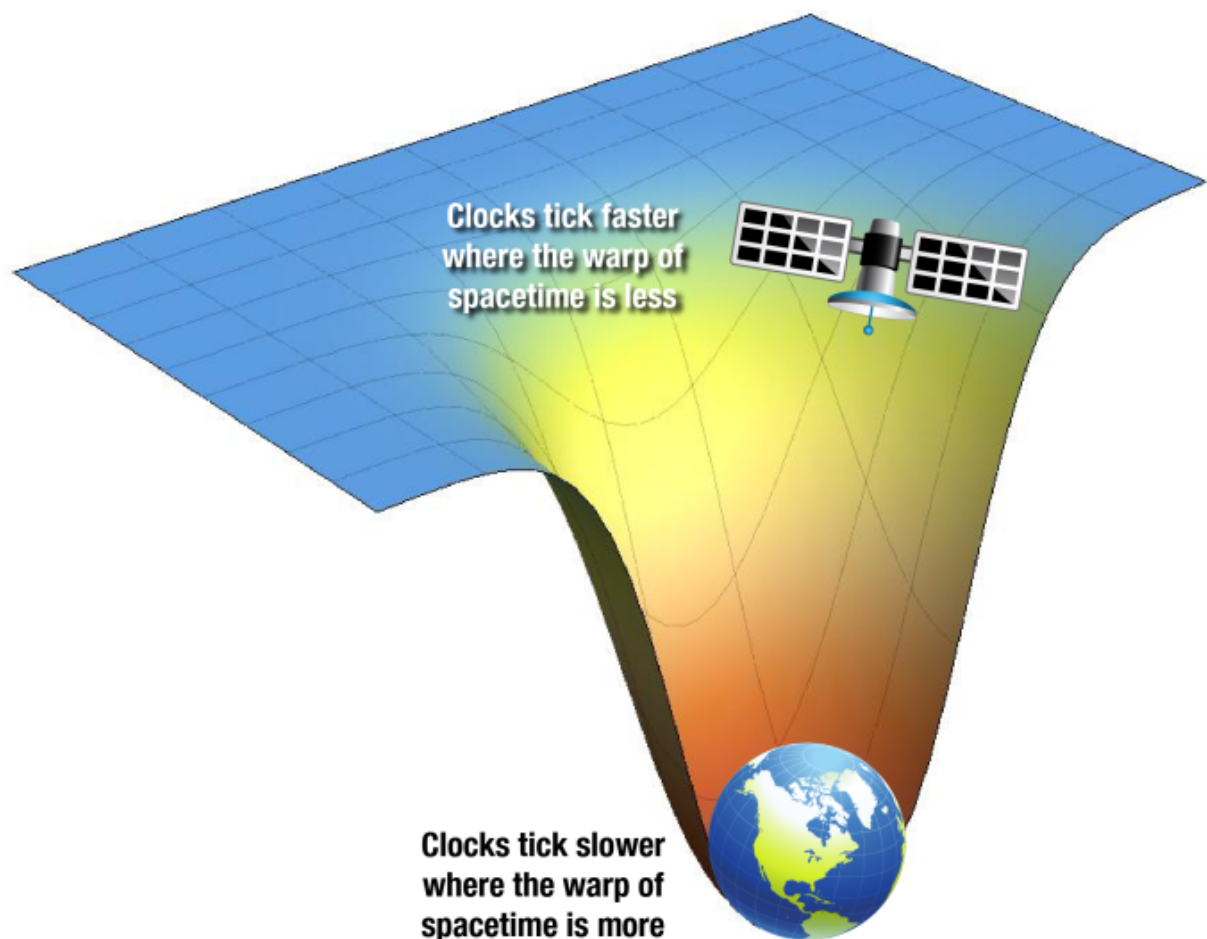
A 2010 experiment showed that identical atomic clocks will keep time slightly differently if one is placed on a higher shelf than the other. (The higher one will gain about a billionth of a second every year, because spacetime is slightly less warped 30cm further from the centre of Earth's mass).

But general relativity has given us much more than a warped view of the universe.

Without the mathematical laws Einstein derived, we wouldn't have any reliable way of predicting or explaining the behaviour and make up of our universe – let alone a reliable GPS.

Still with us? Here are six things that need general relativity to work, and may help you manipulate that mind into understanding the concept.

## 1. GPS



[A graphic showing how General Relativity is vital for GPS to be effective.](#) (Shane L. Larson/Write Science)

GPS relies on satellites orbiting high above us, where Earth's gravity is weaker (because spacetime is less warped by Earth's mass the further away you go).

So the ultra-precise atomic clocks in the satellites run 45 millionths of a second faster per day than clocks here on the ground, deep in Earth's gravitational well.

It's not that the clocks on the satellite are less accurate in space, it's that time is actually passing at a different rate to here on Earth.

If those millionths of a second weren't taken into account when the satellite signals were synced, your GPS coordinates would be out by more than 10 kilometres.

Between that and the annoying voices, GPS would never have taken off.

## 2. Mercury's orbit

Mercury's orbit has always been a little strange — it's got this weird little kink that none of the other planets have.

The only way 19th century mathematicians could account for the kink was if Mercury was being pulled on by something else — like the gravity from another nearby planet, somewhere between Mercury and the Sun.

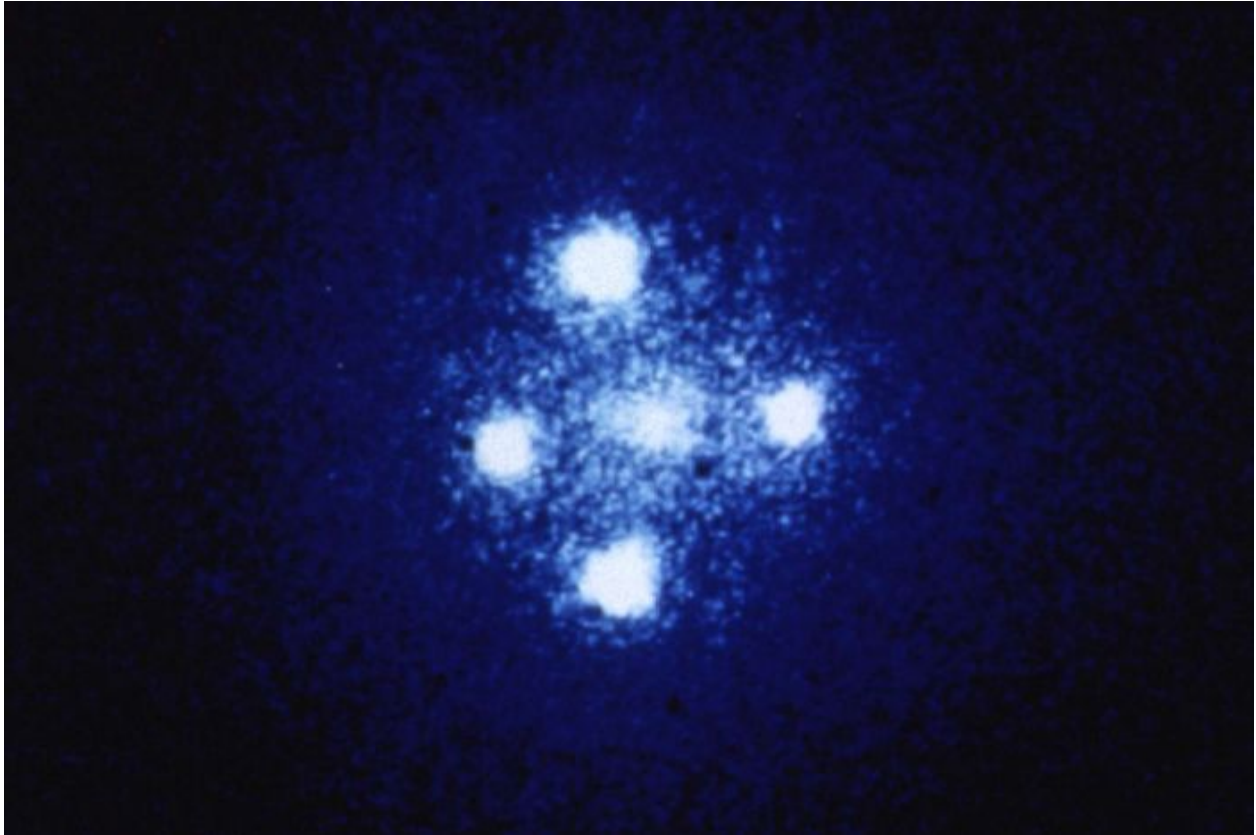
But the proposed planet — Vulcan — was never found. If Vulcan's gravity wasn't the cause, what was? The wrong maths.

Abandoning Newton's theory of gravity, Einstein proposed that Mercury's orbit was odd because it was moving through space that was warped by the Sun's enormous mass.

The effect isn't noticeable on the other planets' orbits because space gets less warped the further you go from the Sun.

General relativity's ability to perfectly account for Mercury's movements was the theory's first test, and it aced it.

### 3. Gravitational lensing



[A photograph taken by NASA's Hubble Space telescope shows gravitational lensing at work - the four stars are actually images of a single quasar which have been multiplied by a nearby galaxy \(in the middle of the image\) acting as a gravitational lens.](#) (NASA, ESA, and STScI)

Warped space doesn't just affect matter and time — it bends light around massive objects too.

The warped space acts like a lens, making things like distant stars and galaxies visible around closer massive objects that should be blocking them from view.

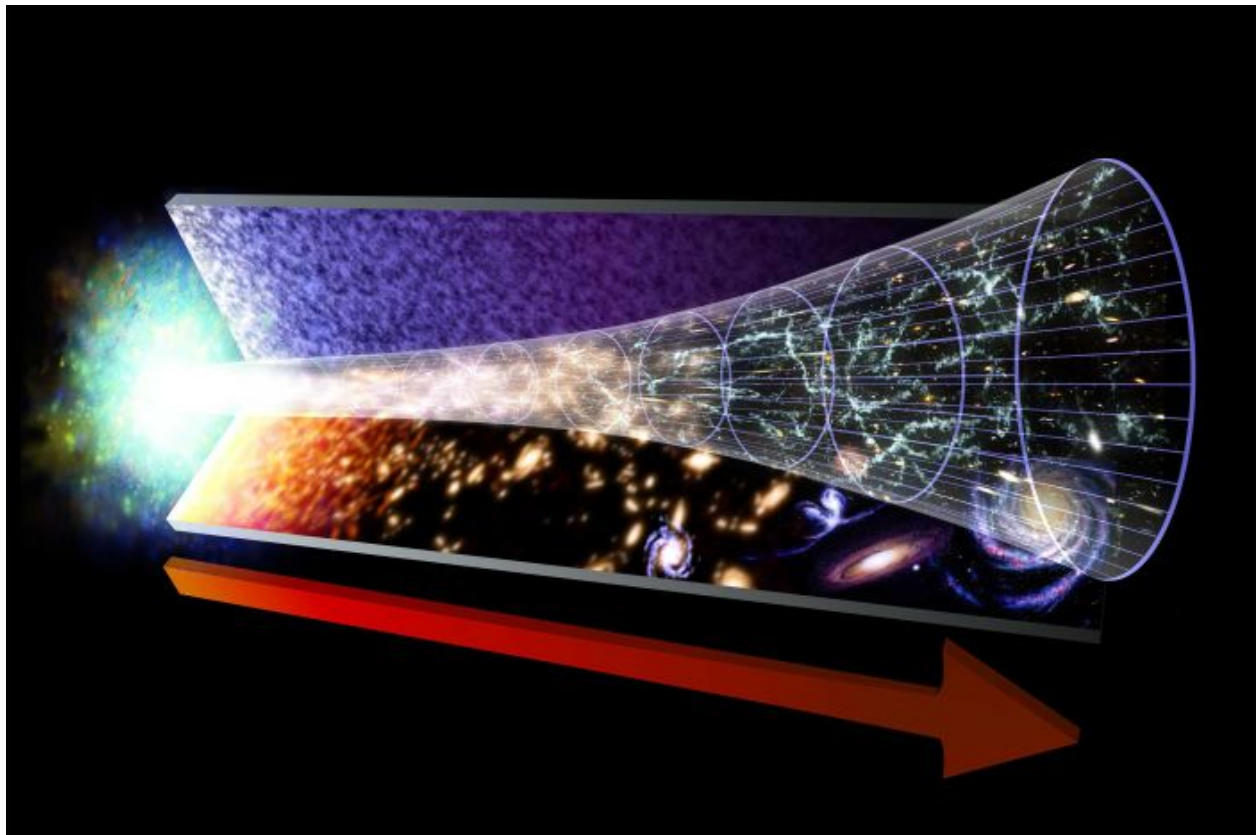
This gravitational lensing was predicted by general relativity — and observations of the lensing of stars near the Sun during eclipses in 1919 and 1922 cemented the theory, making the dashing young Einstein your basic superstar.

Gravitational lensing on a tinier scale (microlensing) was used to find the first Earth-like planets around other stars.

A planet makes a small but noticeable contribution to the gravitational lensing effect of its star on the light from another star behind it.

If both stars are lined up with our observatories on Earth, a planet-sized increase in the magnification of the hidden star can be picked up.

### 4. The Big Bang Theory



[This image represents the evolution of the Universe, starting with the Big Bang. The red arrow marks the flow of time.](#) (NASA)

General relativity says that when spacetime stretches around a massive object, the light travelling through that spacetime stretches too.

So light that starts out one colour ends up a slightly different (longer wavelength) colour after travelling through stretched spacetime.

This colour shift is called cosmological redshift (red is the colour with the longest wavelength) and the extent of redshift in their light lets us measure how far away other galaxies are.

Hubble's observations of cosmological redshift in distant galaxies showed that they were receding away from us at huge speed — which led to the discovery that the universe is expanding.

And Belgian priest/astrophysicist Lemaitre was able to trace those expanding galaxies back to a single point of origin for the universe, a little idea that became known as the Big Bang Theory.

## 5. Black holes

One of the earliest solutions to Einstein's general relativity equations was calculated by German physicist Karl Schwarzschild in 1916, when the theory was still hot off the press

This result gave us one of nature's (and Hollywood's) great blockbusters — black holes.

These infinitely dense remnants of massive dead stars are so tiny, and their gravity so strong, that the escape velocity is faster than the speed of light.

Einstein thought black holes sounded nuts, but there's plenty of indirect evidence that they exist.

## 6. Dark matter





[A composite image showing the galaxy cluster 1E 0657-56, better known as the bullet cluster. Gravitational lensing was used to locate the dark matter \(shown as blue patches\) in these two colliding galaxies. The pink colour shows gas blown apart by the collision.](#) (NASA/Chandra X-Ray Observatory)

Dark matter remains one of the big mysteries of the universe.

In fact, the only thing we know about it is that it has mass.

And we only know that because the gravity resulting from its mass distorts the light coming from galaxies behind it.

General relativity tells us how gravitational lensing should affect this light, and physicists can work backwards to calculate how much dark matter is contributing to the distortion.

## So that's sorted then... well not quite

With general relativity, Einstein gave physicists the maths that let them make predictions about the universe based on how gravity affected spacetime. It turned their ideas about the cosmos into science.

It has passed every test that has been proposed, but there is a problem with the theory.

[It can never work with quantum theory](#), which is extremely successful at predicting and describing the working of the universe at the subatomic level.

Black holes are one of the few places where both general relativity and quantum theory should apply — relativity because of the massive space-warping capability of the dead star, and quantum physics because of the infinitely small size of the singularity at the centre of the black hole.

But that singularity is the problem. General relativity demands an infinitely small point; quantum theory says there's no such thing as points at that scale — wave-like nature means everything at the quantum level is more like a smear.

Both theories can't be right, and right now their inability to be reconciled — or replaced by a testable theory that can work at both the relativistic and quantum levels — leaves a chasm in our physics about

the size of a black hole.

But researchers around the world are working hard to reconcile the two, so hopefully in a century or so we'll be celebrating the anniversary of a unified theory.

Einstein would be chuffed.

*Thanks to Professor David Jamieson at the School of Physics, University of Melbourne*

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